

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Backend & Distributed Systems Aug 2025 to May 2026

- Cut experiment setup effort **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making every experiment sweep reproducible from config.
- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Mentored **200+ students** across **3** courses (Programming Paradigms, AI, ML) through debugging sessions, code reviews, and grading.

Turito | Software Engineering Intern, Backend Reliability Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 CPU **22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Lowered network egress **14%** by removing redundant media-path processing and simplifying routing and payload handling.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, C++17, Rust, SQL

Backend: FastAPI, Flask, REST, gRPC, asyncio, Kafka, Temporal, PostgreSQL, Redis, SQLAlchemy

Security & Reliability: JWT auth, rate limiting, webhook signature verification (6 merged OSS fixes), retries and timeouts, structured metrics

Frontend & Infra: React, TypeScript, Next.js, Docker, AWS, pytest, CI/CD

PROJECTS

Portcullis, Policy Gateway | *Rust*

- Built a **Rust** proxy that centralizes authorization and rate limiting and emits auditable accept, reject, and throttle decisions.

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

Graph Connect, Graph-Backed Platform | *FastAPI, Neo4j, MongoDB, React, Docker, AWS*

- Modeled **50,000+** relationships in **Neo4j** with two-hop Cypher recommendations and load-tested APIs with Locust at about **1,200** users against sub-300 ms targets.

OPEN SOURCE

- **corsairdev/corsair**: 6 merged **webhook signature verification** fixes (Telegram, Teams, SharePoint, Monday, GitLab, Jira) that fail closed
- **google/osv-scalibr**: upstream change to Google's software composition analysis library (open PR, awaiting review)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Database Systems, Distributed Computing, Advanced Algorithms, Computer Networks, Machine Learning, NoSQL

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)